LEARNER GUIDE

Reasoning from Nets

2D and 3D Shapes. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A net is a flat shape that folds up into a solid. In this lesson you move between the two — folding a net into a 3D solid, and reasoning about which flat arrangements will actually close up. You will discover that each face of the net becomes a face of the solid, that opposite faces can be worked out in advance, and the surprising fact that there are exactly eleven different nets of a cube. The question you are exploring is: how does a flat shape fold up into a solid, and which flat shapes will?

What you are learning to notice

- That a net folds into a solid, and each 2D face of the net becomes one face of the 3D solid.
- How to work out which faces will end up opposite each other before you fold.
- That not every arrangement of six squares folds into a cube — there are exactly eleven that do.
- That the area of a net and the surface area of the solid it makes are always the same.

Moving through it, one step at a time

Use the Net Folder: choose a solid and drag the fold slider from flat to folded, then drag the shape to rotate it and see the back. In Predict & Fold, work out the solid and the opposite face before you check. In 'Will it close?', decide which arrangements of six squares fold into a cube. Then design your own net and watch it fold. There is no rush: predict before you check, and use the isometric grid on the Scratchpad to sketch solids.

> **A gentle reminder**
> You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the faces of a solid, naming 2D shapes, and the area of a rectangle), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Learner Guide v0.1.